

Rating Plausibility of Word Senses in Ambiguous Sentences through Narrative Understanding: SemEval 2026 Task 5 — AmbiStory

Shubhadeep Mandal Satyajit Priyadarshi Sangarsu Abhijith
IIIT Hyderabad

{shubhadeep.mandal, satyajit.priyadarshi, sangarsu.abhijith}@students.iiit.ac.in

Abstract

We present our approach to SemEval 2026 Task 5 (AmbiStory): predicting the plausibility of word senses in short ambiguous narratives. We systematically develop eight models, progressing from a single-encoder BiSimpleRNN with random embeddings ($\rho \approx 0.072$) to a dual-encoder BiGRU with KL-divergence annotator-distribution supervision and setup-aware ranking loss ($\rho = 0.640$). The key finding is that treating the task as a comparative within-story ranking problem—rather than independent-sample regression—is the most impactful architectural decision, delivering a +0.327 single-step Spearman gain without any transformer-based architecture.

1 Introduction

The AmbiStory dataset [Gehring and Roth, 2025] presents five-sentence stories where the fourth sentence contains a homonym. The goal is to predict, on a 1–5 scale, how plausible a given word sense is given the full narrative context, as judged by human annotators. Unlike classical word sense disambiguation (WSD), there is no single correct answer: plausibility is graded and reflects genuine human disagreement. This dataset forms the basis of SemEval 2026 Task 5 [Gehring et al., 2026], organised by the NLU Lab at the University of Technology Nuremberg.

This task is challenging because: (1) plausibility is continuous and graded; (2) multi-sentence narrative coherence is required; (3) the dataset is small, increasing overfitting risk; and (4) standard regression losses misalign with the rank-correlation evaluation metric.

We make the following contributions:

- A systematic eight-model progression isolating the contribution of each design choice.
- A stall analysis identifying five root causes of performance plateaus after Models 1–6.

- A task-aware reformulation—treating each story setup as a comparative ranking unit—that more than doubles the prior best Spearman correlation without transformers.
- An annotator-distribution auxiliary objective (KL divergence on 5-way rating distributions) that further improves to $\rho = 0.640$.

2 Background and Related Work

Word Sense Disambiguation. Classical WSD assigns a single, discrete sense to a word in context [Navigli, 2009]. Neural approaches have dominated since Raganato et al. [2017], with pre-trained language models achieving strong performance [Hadiwinoto et al., 2019, Blevins and Zettlemoyer, 2020]. The Word-in-Context (WiC) dataset [Pilehvar and Camacho-Collados, 2019] targets binary sense matching in context. AmbiStory [Gehring and Roth, 2025] differs fundamentally: it requires continuous, graded plausibility scoring over full narratives rather than hard sense assignment.

Graded Lexical Semantics. Erk et al. [2013] introduced graded word meaning, showing human judgments of sense applicability are continuous. McCarthy and Navigli [2009] addressed word sense induction with ranked senses. Gehring and Roth [2025] extend this paradigm to multi-sentence stories, collecting 19,701 human plausibility ratings on 1,899 narratives, forming the basis of SemEval 2026 Task 5 [Gehring et al., 2026].

Semantic Textual Similarity. STS tasks [Cer et al., 2017] also require computing compatibility between two texts. Dual-encoder and cross-encoder architectures have been widely studied for STS [Reimers and Gurevych, 2019, Devlin et al., 2019]. We draw on dual-encoder ideas for our story/meaning comparison.

Learning to Rank. Pairwise and listwise ranking losses have been applied in information retrieval

[Burges et al., 2005] and NLP tasks [Liu, 2009]. We adapt pairwise margin ranking to within-story setup pairs, which is a key contribution of Model 7.

Annotator Disagreement Modeling. Recent work has shown that modeling annotator disagreement as a distribution target rather than a mean label improves calibration and robustness [Uma et al., 2021, Plank, 2022]. Our KL-divergence objective (Model 8) is directly motivated by this literature.

Narrative Understanding. Multi-sentence narrative coherence has been studied in story cloze tasks [Mostafazadeh et al., 2016] and commonsense reasoning [Zellers et al., 2018]. AmbiStory similarly requires tracking causal and temporal information across sentences to determine sense plausibility.

3 Dataset

The AmbiStory dataset [Gehring and Roth, 2025] provides short five-sentence stories for SemEval 2026 Task 5 [Gehring et al., 2026]. The organizers released separate `train.json`, `dev.json`, and `test.json` files for download, and we used these official splits directly without performing any additional split on our end. Each sample contains the following fields:

- **homonym:** the ambiguous target word (e.g., *potential*).
- **judged_meaning:** the candidate word sense being rated.
- **precontext:** three sentences grounding the story.
- **sentence:** the ambiguous fourth sentence containing the homonym.
- **ending** (optional): a fifth sentence often implying a specific sense.
- **choices:** list of five raw integer ratings (1–5) from individual Prolific annotators.
- **average:** mean of choices — the regression target (e.g., 3.0).
- **stdev:** standard deviation of choices, used in the Accuracy within Std Dev metric (e.g., ≈ 1.581).
- **nonsensical:** per-annotator boolean flags indicating whether the annotator found the meaning nonsensical in context.

- **sample_id:** unique identifier linking to the original story setup.
- **example_sentence:** a standalone, context-free usage exemplar of the judged meaning, providing a direct sense anchor independent of the story.

Key statistics. Total story setups: 1,140. Each setup has exactly two judged meanings, yielding 2,280 samples in total. Annotators provided five ratings per sample via the Prolific platform [Gehring and Roth, 2025]. Mean absolute label gap between paired meanings per setup: 1.83. Pairs with gap > 1.0 : 792/1,140 (69.5%). This last statistic is central to our reformulation: the dataset is 69.5% separable by correct within-setup ranking, yet standard regression training never exploits this structure.

Evaluation Metrics. The primary metric is **Spearman rank correlation** (ρ): whether the model’s predicted ordering matches the human ordering. The secondary metric is **Accuracy within Standard Deviation**: the proportion of predictions that fall within $\max(\sigma, 1.0)$ of the mean score.

4 Methodology

We develop eight models in sequence, each addressing specific diagnosed failures of the previous. Table 1 (Section 6) summarizes the full progression; Figures 8 and 9 visualise the metric trajectories.

4.1 Model 1: Single-Encoder BiSimpleRNN

Motivation. Establish the most basic baseline: can a recurrent network with random embeddings predict human plausibility scores from full concatenated text?

Architecture. Figure 1 shows the architecture. All text fields are concatenated into a single string. A trainable embedding layer ($|V| = 10,000$, $d \in \{64, 128, 256\}$) feeds 1–3 stacked Bidirectional SimpleRNN layers, followed by batch normalization, a dense layer with dropout, and a linear output.

Loss & Training. MSE loss, Adam optimizer. Keras Tuner RandomSearch (15 trials).

4.2 Model 2: Dual-Encoder, GloVe Frozen

Motivation. Model 1 has no semantic knowledge and no structural separation between story and meaning. Model 2 addresses both.

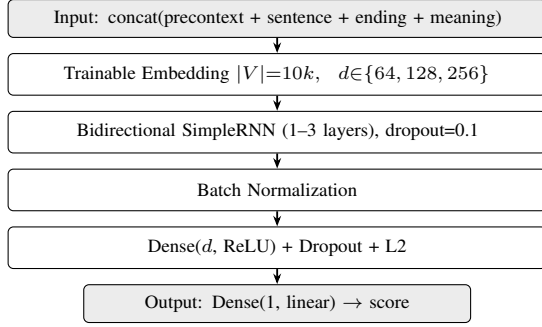


Figure 1: Model 1: Single-Encoder BiSimpleRNN with random embeddings.

Architecture. Figure 2 shows the dual-encoder design. Two independent encoders (identical architecture, separate weights) process the story and meaning streams in parallel. GloVe 6B 100d embeddings are held frozen. Representations are combined via concatenation with absolute difference: $[\mathbf{s}; \mathbf{m}; |\mathbf{s} - \mathbf{m}|]$.

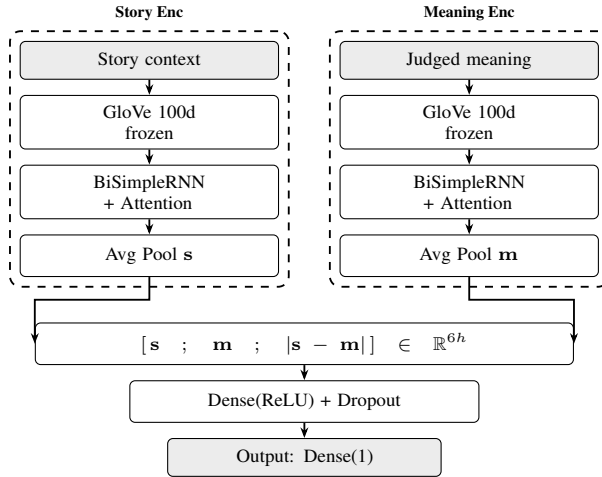


Figure 2: Model 2: Dual-encoder with frozen GloVe and explicit comparison.

4.3 Model 3: Dual-Encoder, GloVe Trainable

Motivation. Test whether allowing GloVe embeddings to fine-tune on the AmbiStory task resolves the residual semantic gap.

Architecture. Identical to Figure 2 (Model 2); the only change is `requires_grad=True` on both embedding layers. Migrated to PyTorch for gradient clipping control (`max_norm=5.0`).

Loss & Training. MSE loss. See Table 2 for a direct comparison of Models 3–5 under identical encoder conditions.

4.4 Model 4: Gaussian Weighted MSE

Motivation. Unweighted MSE treats a near-consensus sample ($\sigma=0.3$) identically to a high-disagreement sample ($\sigma=2.1$). Model 4 weights each sample by $w_i = 1/\sigma_i^2$, and uses `WeightedRandomSampler` to over-sample reliable examples.

Architecture & Loss. Figure 3 shows the encoder stack and loss formulation. The dual BiSimpleRNN encoders and comparison head are unchanged from Model 3; only the loss and sampler differ.

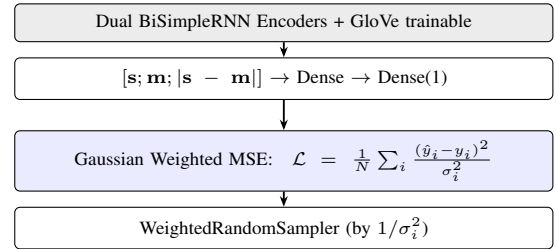


Figure 3: Model 4: Gaussian-weighted MSE with over-sampling of consensus examples.

4.5 Model 5: Huber Loss

Motivation. Replace custom weighted loss with standard robust Huber loss ($\delta = 1.0$), which behaves as MSE for $|\epsilon| \leq \delta$ and as MAE otherwise. This naturally down-weights extreme errors without per-sample weights.

Loss. Figure 4 illustrates the piecewise Huber formulation. The encoder architecture is unchanged from Models 3–4.

$$\mathcal{L}_i^{\text{Huber}} = \begin{cases} \frac{1}{2}(\hat{y}_i - y_i)^2 & |\hat{y}_i - y_i| \leq \delta \\ \delta (|\hat{y}_i - y_i| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

↓

Quadratic ($ \epsilon \leq 1.0$): smooth gradient
Linear ($ \epsilon > 1.0$): clipped gradient

Figure 4: Model 5: Huber loss ($\delta=1.0$). 42.7% of samples fall in the linear zone.

4.6 Model 6: Naive Pairwise Ranking Loss

Motivation. Pointwise losses never directly optimize ranking. Model 6 combines MSE with a pairwise margin ranking loss over within-batch pairs, as shown in Figure 5.

Failure analysis. Model 6 collapses ($\rho = 0.174$, worse than Model 5). Three causes: (1) within-

$$\mathcal{L}_{\text{rank}} = \frac{1}{|P|} \sum_{(i,j) \in P} \max(0, m - (y_i - y_j)(\hat{y}_i - \hat{y}_j))$$

$$\mathcal{L} = 0.5 \cdot \mathcal{L}_{\text{MSE}} + 0.5 \cdot \mathcal{L}_{\text{rank}}, \quad P = \{(i, j) : |y_i - y_j| > 0\}$$

Figure 5: Model 6: Combined MSE + pairwise ranking loss. Collapsed: 99.2% of pair constraints violated.

batch pairs include many near-tied labels generating contradictory gradients; (2) margin=1.0 is too large for the 1–5 scale with high human variance; (3) the SimpleRNN encoders produce insufficiently discriminative vectors for ranking to function.

4.7 Model 7: Breakthrough — BiGRU + Rich Interactions + Meta Features + Setup-Aware Ranking

Motivation. A systematic stall analysis (Section 5) identified five root causes of the $\rho \approx 0.20$ –0.24 plateau. Model 7 applies all five solutions simultaneously.

Architecture. Figure 6 shows the complete architecture.

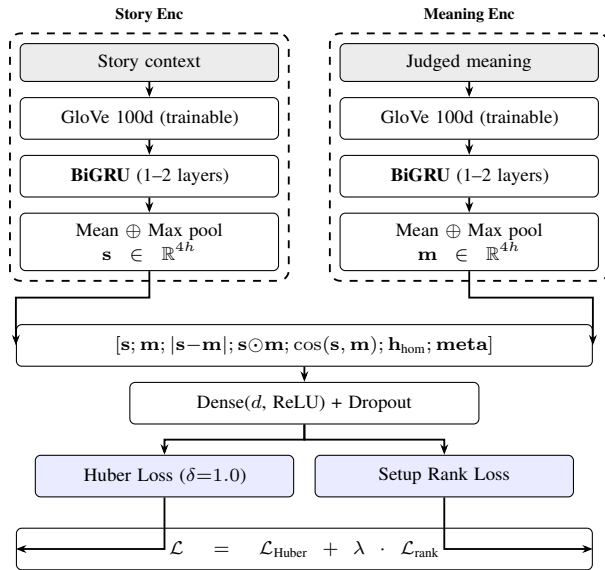


Figure 6: Model 7: BiGRU dual encoder, rich 4-way comparison, homonym/meta features, setup-aware ranking loss. $\rho = 0.572$.

Setup-Aware Ranking Loss. Unlike Model 6’s naive within-batch pairing, the ranking loss is computed only between the two meanings of the *same story setup*. A SetupBatchSampler ensures each batch always contains grouped setup pairs.

$$\mathcal{L}_{\text{rank}} = \frac{1}{|G|} \sum_{g \in G} \frac{1}{|P_g|} \sum_{(i,j) \in P_g} \max(0, m - \text{sgn}(y_i - y_j)(\hat{y}_i - \hat{y}_j)) \quad (1)$$

where G is the set of setups in a batch, P_g is all pairs with differing labels in setup g , and margin m is Optuna-tuned.

Metric-Aligned Optuna. All prior HP search optimized regression loss. Model 7 optimizes:

$$\text{obj} = \rho + 0.25 \cdot \text{AccStd} - 0.05 \cdot \text{MAE} \quad (2)$$

with 35 trials, TPE sampler, and MedianPruner.

4.8 Model 8: BiGRU + KL Annotator-Distribution Head

Motivation. Model 7 supervises only on the mean score. The full 5-way annotator distribution (choices) encodes uncertainty and disagreement that could help calibration.

Architecture. Figure 7 shows the dual-head extension of Model 7.

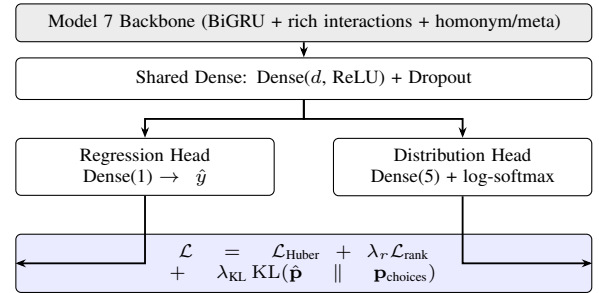


Figure 7: Model 8: Dual output head — scalar regression + 5-way annotator distribution via KL divergence. $\rho = 0.640$.

5 Stall Analysis

After Models 4–6, Spearman plateaued at $\rho \approx 0.20$ –0.24. We identify five root causes. Model 7 (Figure 6) addresses all five simultaneously; Table 3 summarises the evidence for each resolution.

P1: Task Formulation Mismatch. Each story has exactly two candidate meanings (Table ??). The mean within-setup label gap is 1.83; 69.5% of pairs are separable by correct ranking. Models 1–6 never exploited this structure.

P2: Loss–Metric Misalignment in HP Tuning. Prior Optuna runs optimized Huber loss, not Spearman. Equation (2) corrects this.

#	Model	ρ	Acc/Std	MAE
1	Single-Enc, rand emb, MSE	0.072	0.547	1.050
2	Dual-Enc, GloVe frozen	0.021	0.570	1.041
3	Dual-Enc, GloVe trainable	0.186	0.570	1.040
4	Gaussian weighted MSE	0.237	0.617	1.011
5	Huber loss ($\delta=1$)	0.245	0.608	0.993
6	Naive pairwise rank (collapsed)	0.174	0.541	1.036
7	BiGRU + rich + setup rank	0.572	0.725	0.867
8	BiGRU + KL dist head	0.640	0.772	0.748

Table 1: All eight models. Spearman ρ is the primary metric.

Loss	ρ	Acc/Std	MAE
MSE (Model 3)	0.186	0.570	1.040
Gaussian Wt (Model 4)	0.237	0.617	1.011
Huber $\delta=1$ (M5)	0.245	0.608	0.993

Table 2: Loss function comparison on identical BiSimpleRNN encoders.

P3: Unused Dataset Signals. The following fields were completely ignored in Models 1–6: choices (the 5-way raw rating list, richer than average alone); nonsensical (per-annotator boolean flags indicating the meaning was found nonsensical — the count of true values is a direct low-plausibility signal); homonym (the identity of the ambiguous word, which predicts the plausibility range); and an ending-present binary flag (endings frequently bias interpretation toward one sense).

P4: Weak Representation Strategy. Global average pooling discards peak-activation features. The comparison head (see Figure 2) used only $|\mathbf{s} - \mathbf{m}|$, missing element-wise product and cosine similarity — both added in Model 7 (Figure 6).

P5: Validation Leakage. Random splitting can place both meanings of a setup across train/val. GroupShuffleSplit by setup ID is required for realistic validation estimates.

6 Experiments and Results

6.1 Results Summary

Table 1 summarizes all eight models. Figures 8 and 9 show the Spearman and Accuracy/Std progression.

6.2 Loss Engineering Analysis (Models 3–5)

Table 2 compares the three loss variants on identical BiSimpleRNN encoders.

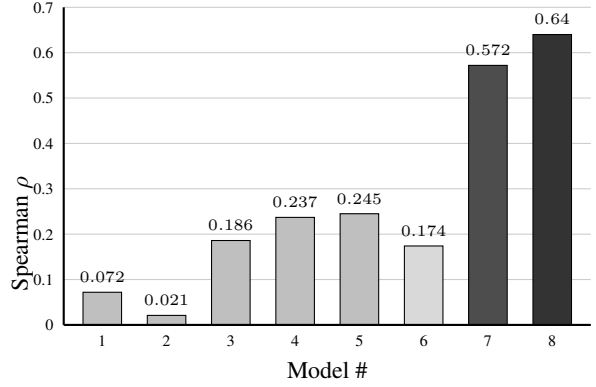


Figure 8: Spearman ρ progression across eight models. Dark bars = Models 7–8 (breakthrough).

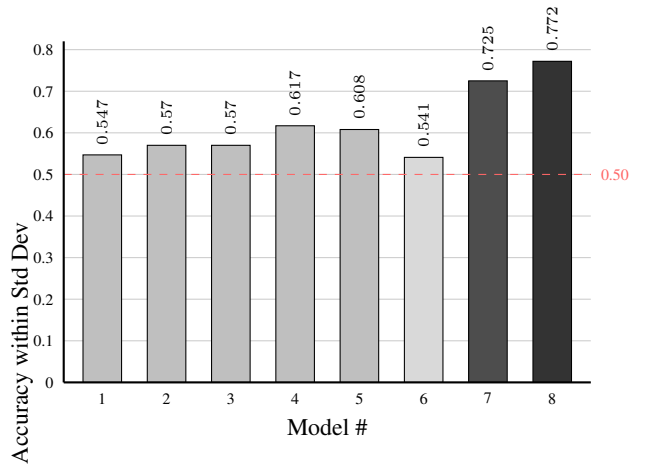


Figure 9: Accuracy within Std Dev across eight models. Dashed red line at 0.50 for reference. Models 7–8 exceed 72%.

Loss engineering extends gains beyond trainable GloVe (Model 3 $\rightarrow$ 5), but with diminishing returns. The Huber–Gaussian difference (+0.008 Spearman) is within typical variance, confirming the bottleneck had shifted to encoder capacity and task formulation.

6.3 Key Transitions

- **Models 1 $\rightarrow$ 2 $\rightarrow$ 3:** Structural separation and trainable GloVe both necessary. M3: $\rho = 0.186$.
- **Models 3 $\rightarrow$ 4 $\rightarrow$ 5:** Loss engineering pushed to $\rho = 0.245$, diminishing returns confirmed.
- **Model 6:** Controlled failure—naive ranking without setup awareness collapsed; provided precise diagnosis.
- **Model 7:** All stall-analysis solutions applied simultaneously. $\rho : 0.245 \rightarrow 0.572$ (+0.327,

Root Cause	Resolution	Evidence
No semantic knowledge	GloVe trainable (M3)	ρ : 0.021→0.186
MSE collapse	Huber/Gauss	ρ : 0.186→0.245
SimpleRNN ceiling	BiGRU (M7)	+0.327 gain
Shallow comparison	Rich $\odot$, cos head:	+0.327 gain
Unused signals	Hom. emb, meta	+0.327 gain
Validation leakage	GroupShuffle-Split	Reliable val. est.
HP obj. mismatch	Metric-aligned HP	+0.327 gain
Annot. distribution	KL head (M8)	ρ : 0.572→0.640

Table 3: Resolved root causes and evidence.

largest single-step gain).

- **Model 8:** KL annotator-distribution auxiliary. ρ : 0.572 → 0.640 (+0.068).

7 Resolved and Remaining Bottlenecks

Table 3 summarises all resolved root causes with supporting evidence. The breakthroughs are concentrated in Model 7 (Figure 6) and Model 8 (Figure 7); full metric progression is in Table 1.

Remaining bottlenecks. BiGRU sequential ceiling (target: BiLSTM then BERT/RoBERTa fine-tuning, Model 10); token-level interaction missing (cross-attention, Model 9, Figure 6 shows where cross-attention will be inserted between encoders); extreme-score weakness (scores 1 and 5 accuracy still low, see Figure 9).

8 Progress and Timeline

8.1 Progress Summary

Completed. Models 1–8 fully implemented and evaluated. Systematic stall analysis documented. Setup-aware ranking loss, metric-aligned Optuna, grouped cross-validation, and KL auxiliary objective all in production.

Ongoing. Ablation study to isolate the contribution of each Model 7 innovation. Post-hoc isotonic calibration of validation predictions. Per-bucket accuracy analysis for Model 8.

Planned — Future Work. Four concrete directions are prioritised for the next phase. **(1) BiLSTM encoders.** We plan to replace the BiGRU encoders in Models 7–8 with Bidirectional LSTM encoders. BiLSTM’s separate input, forget, and output gates give it finer control over

long-range dependencies compared to GRU’s single reset/update gate pair, which may better capture the multi-sentence causal and temporal coherence required for narrative plausibility judgement. **(2) Temperature-scaled softmax for extreme scores.** Models 1–8 consistently underpredict scores of 1 and 5. We will apply temperature scaling to the output softmax during inference, lowering the temperature to sharpen the predicted distribution and encourage the model to commit to extreme plausibility values rather than defaulting to mid-range predictions. **(3) ELMo embeddings with example_sentence.** The dataset provides an `example_sentence` field for each judged meaning, which gives a standalone, context-free usage of the sense. We plan to encode this field using ELMo [Peters et al., 2018] and concatenate the resulting contextualised embedding to the meaning encoder, providing a richer and sense-specific lexical signal beyond GloVe. **(4) Cross-attention and transformer fine-tuning.** Model 9 will introduce cross-attention between story and meaning token sequences (target $\rho > 0.65$); Model 10 will fine-tune BERT/RoBERTa with a sentence-pair head (target $\rho > 0.72$).

8.2 Tentative Timeline

Milestone	Week	Status
Models 1–6 (baselines)	Week 1–3	Done
Stall analysis	Week 4	Done
Model 7 (BiGRU breakthrough)	Week 5	Done
Model 8 (KL head)	Week 5	Done
Mid-report submission	Week 5	Submitted
Ablation study	Week 6–7	In progress
BiLSTM encoder replacement	Week 8	Planned
Temp. scaling (extreme scores)	Week 8	Planned
ELMo + <code>example_sentence</code>	Week 9	Planned
Model 9 (cross-attention)	Week 10	Planned
Model 10 (BERT fine-tuning)	Week 10–11	Planned
System paper + submission	Week 11	Planned

Table 4: Project timeline (11 weeks. Weeks 1–5 complete; 6 weeks remaining).

9 Conclusion

We have developed eight models for SemEval 2026 Task 5 [Gehring et al., 2026], progressing from $\rho = 0.072$ to $\rho = 0.640$. The central scientific finding is that treating word sense plausibility as independent-sample regression fundamentally misaligns with the task structure. Reformulating each story as a comparative within-setup ranking problem—with

setup-grouped batching, setup-aware ranking loss, and metric-aligned HP search—delivered the single largest performance gain in the series (+0.327) without any transformer architecture. An additional +0.068 was obtained by incorporating annotator rating distributions as KL-divergence supervision (Model 8).

The next phase (Table 4) targets four improvements: (1) replacing BiGRU with BiLSTM encoders for finer long-range gating; (2) temperature-scaled softmax to sharpen predictions at extreme plausibility scores (1 and 5); (3) ELMo-encoded `example_sentence` features [Peters et al., 2018] to enrich the meaning encoder with context-free sense exemplars from the dataset; and (4) cross-attention followed by BERT/roBERTa fine-tuning, targeting $\rho > 0.72$.

References

- Janosch Gehring and Michael Roth. 2025. AmbiStory: A challenging dataset of lexically ambiguous short stories. In *Proceedings of the 14th Joint Conference on Lexical and Computational Semantics (*SEM 2025)*, pages 152–171, Suzhou, China. Association for Computational Linguistics.
- Janosch Gehring, Selina Meyer, and Michael Roth. 2026. SemEval 2026 Task 5: Rating plausibility of word senses in ambiguous sentences through narrative understanding. In *Proceedings of SemEval 2026*. Association for Computational Linguistics. <https://nlu-lab.github.io/semeval.html>
- Roberto Navigli. 2009. Word sense disambiguation: A survey. *ACM Computing Surveys*, 41(2):1–69.
- Alessandro Raganato, Jose Camacho-Collados, and Roberto Navigli. 2017. Word sense disambiguation: A unified evaluation framework and empirical comparison. In *Proceedings of EACL 2017*, pages 99–110.
- Christian Hadiwinoto, Hwee Tou Ng, and Wee Chung Gan. 2019. Improved word sense disambiguation using pre-trained contextualized word representations. In *Proceedings of EMNLP-IJCNLP 2019*.
- Terra Blevins and Luke Zettlemoyer. 2020. Moving down the long tail of word sense disambiguation with gloss informed bi-encoders. In *Proceedings of ACL 2020*, pages 1006–1017.
- Mohammad Taher Pilehvar and Jose Camacho-Collados. 2019. WiC: The word-in-context dataset for evaluating context-sensitive meaning representations. In *Proceedings of NAACL-HLT 2019*, pages 1267–1273.
- Katrin Erk, Diana McCarthy, and Nicholas Gaylord. 2013. Measuring word meaning in context. *Computational Linguistics*, 39(3):511–554.
- Diana McCarthy and Roberto Navigli. 2009. The English lexical substitution task. *Language Resources and Evaluation*, 43(2):139–159.
- Daniel Cer, Mona Diab, Eneko Agirre, Inigo Lopez-Gazpio, and Lucia Specia. 2017. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *Proceedings of SemEval-2017*, pages 1–14.
- Nils Reimers and Iryna Gurevych. 2019. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of EMNLP-IJCNLP 2019*, pages 3982–3992.
- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT 2019*, pages 4171–4186.
- Chris Burges, Tal Shaked, Erin Renshaw, Ari Lazier, Matt Deeds, Nicole Hamilton, and Greg Hullender. 2005. Learning to rank using gradient descent. In *Proceedings of ICML 2005*.
- Tie-Yan Liu. 2009. Learning to rank for information retrieval. *Foundations and Trends in Information Retrieval*, 3(3):225–331.
- Alexandra Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research*, 72:1385–1470.
- Barbara Plank. 2022. The “problem” of human label variation: On ground truth in data, modeling and evaluation. In *Proceedings of EMNLP 2022*.
- Nasrin Mostafazadeh, Nathanael Chambers, Xiaodong He, Devi Parikh, Dhruv Batra, Lucy Vanderwende, Pushmeet Kohli, and James Allen. 2016. A corpus and cloze evaluation for deeper understanding of commonsense stories. In *Proceedings of NAACL-HLT 2016*, pages 839–849.
- Rowan Zellers, Yonatan Bisk, Roy Schwartz, and Yejin Choi. 2018. SWAG: A large-scale adversarial dataset for grounded commonsense inference. In *Proceedings of EMNLP 2018*, pages 93–104.
- Matthew Peters, Mark Neumann, Mohit Iyyer, Matt Gardner, Christopher Clark, Kenton Lee, and Luke Zettlemoyer. 2018. Deep contextualized word representations. In *Proceedings of NAACL-HLT 2018*, pages 2227–2237. Association for Computational Linguistics.